

# Technical note: ASTRA weighted correlation functions — replacing environment *splits* with a continuous *weight*

ASTRA clustering project

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## Abstract

The ASTRA analysis classifies each object by a local cosmic-web parameter  $r = (n_{\text{data}} - n_{\text{rand}})/(n_{\text{data}} + n_{\text{rand}}) \in [-1, 1]$  obtained from a Delaunay tessellation over the joint data+random catalog, then splits the sample into quantiles of  $r$  and measures a two-point correlation function (2PCF) per quantile. Here we develop an alternative that uses  $r$  directly as a per-object *weight* and measures *weighted* (marked) correlation functions, avoiding the information loss and bin-edge discontinuities of the split. We are careful to keep three populations distinct: the *data* galaxies (which carry redshift-space distortions), the *ASTRA randoms* (the  $1\times$  uniform points that enter the Delaunay and therefore carry an  $r$ ), and the *geometry randoms* (the  $5\times$  uniform catalog that is only the Landy–Szalay reference). Crucially, only the two  $r$ -carrying populations are weighted; the geometry randoms define the window and are never weighted. We present the estimator (a unified weighted Landy–Szalay with mean-normalised weights, so the uniform weight reproduces the standard 2PCF exactly), seven weight schemes, three statistics (data auto, ASTRA-random auto, and their cross), and the first full-box measurement on the fiducial c000/hod484 box. This note documents the method in full; the results section is populated as the first run completes.

## 1 Motivation

The quantile split has two practical drawbacks. First, it discards information: the continuous  $r$  is reduced to a four-level label, and the bin edges are arbitrary. Second, it is not smooth: the bin boundaries introduce discontinuities that are awkward for the emulator/Fisher derivatives that the rest of the project relies on, and they enlarge the data vector (four quantiles  $\times$  several leg families), inflating the Hartlap penalty. A continuous weight  $w = f(r)$  keeps the full  $r$  information, is differentiable in the cosmological and HOD parameters, and produces a compact data vector. Weighted (“marked”) correlation functions are an established tool in large-scale structure with known sensitivity to modified gravity, massive neutrinos, and assembly bias [1, 2, 3], which strengthens the science case.

## 2 Three populations, and which ones are weighted

The pipeline keeps three populations strictly separate; conflating them is the main pitfall.

- **Data** ( $N_{\text{d}}$  galaxies): carry redshift-space distortions (the line-of-sight coordinate is `Z_RSD`); each has an  $r$ .
- **ASTRA randoms** ( $1\times$  data; regenerated each iteration with seed `SEED+iter`): the uniform points that enter the Delaunay tessellation, so each carries an  $r$ . These are the “random quantile” tracers of the split analysis.

- **Geometry randoms** ( $5\times$  data; fixed seed): a uniform catalog used *only* as the Landy–Szalay reference (the survey window). They do *not* enter the Delaunay and therefore have no  $r$ .

Because the geometry randoms have no  $r$ , they cannot and must not be weighted by  $f(r)$ : weighting the LS reference by a data-derived quantity would distort the window rather than the signal. The weights attach only to the data and the ASTRA randoms; the geometry randoms remain uniform and unweighted. This is the continuous analog of the split analysis, in which the data and the ASTRA randoms are each binned, while the geometry randoms provide the common reference for every 2PCF.

### 3 Estimator

#### 3.1 The environment parameter

For each object in the joint catalog, the Delaunay neighbour counts give

$$r = \frac{n_{\text{data}} - n_{\text{rand}}}{n_{\text{data}} + n_{\text{rand}}} \in [-1, 1], \quad (1)$$

with  $r \rightarrow -1$  in voids and  $r \rightarrow +1$  in knots. Objects with no neighbours ( $n_{\text{data}} + n_{\text{rand}} = 0$ , exceedingly rare) have  $r$  undefined and are dropped uniformly across all schemes, so every scheme uses the same sample.

#### 3.2 Weighted Landy–Szalay with mean-normalised weights

Given a per-object mark  $m_i = f(r_i)$  for a tracer population, we form mean-normalised weights

$$w_i = \frac{f(r_i)}{\langle f(r) \rangle}, \quad \langle w \rangle = 1, \quad (2)$$

computed separately per population and per ASTRA iteration. The weighted 2PCF is the standard Landy–Szalay estimator with these weights on the tracer(s) and unit weights on the geometry randoms  $G$ :

$$\hat{\xi}_w(s, \mu) = \frac{D_w D_w - 2 D_w G + G G}{G G} \quad (3)$$

for an auto-correlation, and the analogous cross form  $\hat{\xi}_w = (D_w A_w - D_w G - A_w G + G G)/G G$  for the data $\times$ ASTRA-random cross, where  $A_w$  denotes the weighted ASTRA randoms. Pair counts are normalised by the sums of weights, which `pycorr` handles natively. The geometry-random pair count  $G G$  is computed once and reused by every estimator. The multipoles  $\xi_{w,\ell}(s)$  for  $\ell = 0, 2$  follow by the usual Legendre projection.

The mean normalisation (2) has a useful consequence: the *uniform* scheme  $f \equiv 1$  gives  $w_i \equiv 1$  and (3) reduces to the ordinary 2PCF. We use this as a built-in unit test (the uniform weighted curve must coincide with the standard full-data  $\xi$ ).

#### 3.3 Marked correlation for signed marks

For the signed mark  $m = r$  (which changes sign between voids and knots), the mean-normalised weighted estimator measures the marked-correlation *numerator*  $W(s)$ . The marked correlation function itself is the ratio

$$M(s) = \frac{1 + W(s)}{1 + \xi(s)}, \quad (4)$$

with  $\xi$  the unmarked (uniform) correlation;  $M(s) = 1$  is the null of no mark–clustering correlation,  $M > 1$  that high- $r$  objects cluster more strongly at scale  $s$ . We report  $M_0(s)$  formed per iteration from the saved  $W_0$  (signed) and  $\xi_0$  (uniform) arrays.

## 4 Weight schemes

We measure the following  $f(r)$  (Table 1). All but the signed mark are non-negative, so their weighted  $\xi_{w,\ell}$  are clean Landy–Szalay objects; the signed mark is reported through the marked-correlation ratio (4).

| Scheme  | $f(r)$             | character                                          |
|---------|--------------------|----------------------------------------------------|
| uniform | 1                  | baseline; reproduces the standard 2PCF (unit test) |
| knot    | $(1 + r)/2$        | overdense emphasis (continuous Q4)                 |
| void    | $(1 - r)/2$        | underdense emphasis (continuous Q1)                |
| rank    | $\text{rank}(r)/N$ | distribution-robust continuous quantile            |
| power   | $[(1 + r)/2]^2$    | sharpened knot emphasis                            |
| exp     | $\exp(2r)$         | smooth, tunable, positive                          |
| signed  | $r$                | signed density weighting; marked correlation $M_0$ |

Table 1: Weight schemes. The rank is computed within each population (data, ASTRA randoms) separately, per iteration.

## 5 Statistics

For each scheme we measure the weighted multipoles  $\ell = 0, 2$  of three statistics:

- **data auto:** data weighted by  $f(r_{\text{data}})$ ;
- **ASTRA-random auto:** ASTRA randoms weighted by  $f(r_{\text{A}})$ ;
- **data  $\times$  ASTRA-random cross:** the two weighted fields of the *same* iteration, cross-correlated.

The cross quadrupole is expected to be the most informative of the three: the data carry RSD while the ASTRA randoms do not, so the cross  $\ell = 2$  probes the data’s velocity anisotropy modulated by the environment weighting, whereas the ASTRA-random auto  $\ell = 2$  is expected to be weak (those points have no peculiar velocity).

## 6 Implementation

`scripts/pipeline_fullbox_weighted.py` runs the full  $2h^{-1}$  Mpc Gpc box. Per ASTRA iteration it (i) generates the ASTRA randoms, (ii) builds the joint catalog and runs `astra.classify_fast` to obtain  $r$  for every object, (iii) aligns  $r$  to the data and ASTRA-random position arrays and *stores* them (`fbw_rvalues_iter{it}.npz, float32`) so the weighting can be recomputed or extended without re-running the (expensive) Delaunay, and (iv) computes the weighted multipoles of the three statistics for all seven schemes. Outputs go to `data/fullbox_weighted/{cosmo}_hod{NNN}/` with prefix `fbw_`. `scripts/plot_fullbox_weighted.py` produces the figures; `queue/run_fullbox_weighted.sh`

is the CPU-node wrapper. The `pycorr` weighted API and the uniform-equals-unweighted identity were validated in a smoke test before the first run.

## 7 First test

- Catalog: `c000/hod484` (the Fisher/emulator fiducial), full box.
- ASTRA iterations: 3; display as iteration mean  $\pm$  std.
- Binning:  $s \in [0, 150] h^{-1} \text{ Mpc}$ , 15 bins;  $\mu$  in 240 bins;  $\ell = 0, 2$ .
- Schemes: all seven of Table 1.

## 8 Results

The first run (55159247, `c000/hod484`, 3 iterations, 40 min wall) completed on 2026-06-27 and produced the two planned figures:

- `weighted_xi_multipoles.png`:  $s^2 \xi_{w,\ell}(s)$  for the positive schemes,  $3 \times 2$  (data / ASTRA-random / cross  $\times \ell = 0, 2$ );
- `marked_correlation_signed.png`:  $M_0(s)$  for the signed mark, per statistic.

Representative monopole/quadrupole amplitudes at the bin  $s \simeq 75 h^{-1} \text{ Mpc}$  (index 7) are collected in Table 2.

| scheme  | data $\xi_0$ | arand $\xi_0$ | cross $\xi_0$ | data $\xi_2$ |
|---------|--------------|---------------|---------------|--------------|
| uniform | +0.0021      | $\sim 0$      | $\sim 0$      | -0.0089      |
| void    | +0.0005      | +0.0004       | -0.0005       | -0.0023      |
| knot    | +0.0045      | +0.0006       | +0.0017       | -0.0185      |
| rank    | +0.0071      | +0.0015       | +0.0033       | -0.0285      |
| power   | +0.0071      | +0.0021       | +0.0038       | -0.0287      |
| exp     | +0.0078      | +0.0021       | +0.0040       | -0.0319      |
| signed  | +0.2127      | +0.0405       | -0.0938       | -0.8178      |

Table 2: Weighted multipoles at  $s \simeq 75 h^{-1} \text{ Mpc}$  (bin index 7) for the first test.

The four planned checks all pass: (i) the `uniform` data-auto reproduces the standard full-data  $\xi$ , and its ASTRA-random auto and cross collapse to  $\sim 0$  (uniform weights on uniform points), validating the mean-normalised estimator and unit test; (ii) the schemes *monotonically* bracket the uniform curve — `void` suppresses ( $\xi_0 = +0.0005$ ) and `knot/power/exp` amplify ( $+0.0045 \rightarrow +0.0078$ ) both monopole and quadrupole, the overdense-emphasis ordering expected of an environment mark; (iii) the cross quadrupole is non-zero and grows with the overdense weighting (RSD signal modulated by environment) while the ASTRA-random auto quadrupole stays weak; (iv) the signed marked correlation departs strongly from the uniform null ( $W_0 = +0.21$ ,  $W_2 = -0.82$ , and the cross flips sign), i.e. the environment correlates with clustering at high significance. No binning artefacts or estimator pathologies appeared. The continuous mark therefore responds to environment exactly as the quantile splits do, with the full  $r$  information retained and no bin-edge discontinuities.

## 9 Minimal multi-cosmology Fisher comparison

The first test fixes the estimator; the question that matters is whether the continuous mark carries *cosmological* information competitive with the quantile legs. We answer it on the existing nine-cosmology matched-HOD Fisher grid (the same  $\pm$  pairs that drive the quantile Fisher:  $c100/179, c101/152$  for  $\omega_b$ ;  $c102/556, c103/861$  for  $\omega_c$ ;  $c104/498, c105/589$  for  $n_s$ ;  $c112/507, c113/483$  for  $\sigma_8$ ;  $c000/484$  fiducial). The only new compute is the eight  $\pm$  weighted runs; the covariance is built by *reanalysis* of the cached  $c000$  box (`weighted_subbox_cov.py`: tile into 64 subboxes  $\times 3$  iterations, weighted Landy–Szalay per subbox/scheme,  $C/64$  to full-box volume — no new simulations). For each scheme we form the central-difference derivative of the 90-bin weighted vector (data/arand/cross  $\times \ell=0, 2 \times 15$  bins) and the HOD-fixed marginalised  $\sigma$  on  $\{\omega_b, \omega_c, n_s, \ln \sigma_8\}$ , against the quantile full-sample auto (30 bins,  $C_{\text{subbox}}/64$ ).

Table 3: HOD-fixed marginalised  $\sigma$  at full-box ( $2000 h^{-1}\text{Mpc}$ ) volume, and the gain (quantile-full-auto  $\sigma$  / weighted  $\sigma$ ;  $> 1$  = weighted tighter). Hartlap 0.52 (90 bins, 192 pooled cov samples); quantile reference Hartlap 0.51.

| scheme             | $\omega_b$                      | $\omega_c$                      | $n_s$                           | $\ln \sigma_8$                  |
|--------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
| quantile full-auto | $8.77 \times 10^{-3}$           | $3.11 \times 10^{-3}$           | $1.28 \times 10^{-3}$           | $6.72 \times 10^{-3}$           |
| rank (CDF)         | $1.57 \times$                   | <b><math>1.87 \times</math></b> | <b><math>1.78 \times</math></b> | $1.34 \times$                   |
| power              | <b><math>1.67 \times</math></b> | $1.60 \times$                   | $1.62 \times$                   | $1.28 \times$                   |
| knot               | $1.52 \times$                   | $1.77 \times$                   | $1.72 \times$                   | $1.20 \times$                   |
| uniform            | $1.57 \times$                   | $1.60 \times$                   | $1.65 \times$                   | $1.26 \times$                   |
| exp                | $1.55 \times$                   | $1.43 \times$                   | $1.44 \times$                   | $1.26 \times$                   |
| void               | $1.37 \times$                   | $1.50 \times$                   | $1.43 \times$                   | $1.28 \times$                   |
| signed mark        | <b><math>2.03 \times</math></b> | $0.88 \times$                   | $0.84 \times$                   | <b><math>1.68 \times</math></b> |

Findings: (i) *every positive scheme tightens all four parameters* ( $1.2\text{--}1.9\times$ ) over the quantile full-auto, at a benign Hartlap factor — not a binning artefact. (ii) **rank** (the CDF mark) is the best-balanced scheme, strongest on  $\omega_c$  and  $n_s$ . (iii) The **signed** marked correlation is the specialist: best on  $\omega_b$  ( $2.03\times$ ) and  $\sigma_8$  ( $1.68\times$ ) — the amplitude/RSD-sensitive parameters — but *worse* than the quantile on  $\omega_c/n_s$ , whose constraint leans on the broadband shape the zero-mean mark discards.

Two honest caveats keep these as *relative* numbers. The weighted vector bundles three legs (data-auto, ASTRA-random auto, cross; 90 bins) against the single quantile leg (30 bins), so part of the gain is “more legs,” not the weighting alone; a fairer baseline is the best quantile *multi*-leg vector. And the  $\pm$  pairs use matched-but-mismatched single HODs, so every derivative here carries the same HOD contamination the quantile Fisher had before the Tier-1 / global-response correction (the  $\sigma_8$  row worst). The absolute  $\sigma$  are therefore optimistic; the head-to-head ratios are the result. The HOD-clean upgrade is the weighted analogue on the 50-draw HOD ensemble per cosmology with the global-response derivatives.

## 10 Per-leg breakdown and a fairer quantile baseline

The two caveats above are exactly what `fisher_weighted_breakdown.py` tests, from cached data alone. The conclusion overturns the preliminary headline.

**Re-weighting one population adds nothing.** The cleanest controlled experiment: the **uniform** data-auto *is* the standard  $\xi$ , so re-weighting the same data population by any mark and re-measuring

its auto isolates the pure effect of the weight. No scheme improves on standard  $\xi$  on the clean parameters — knot/rank/power/exp are all 0.75–0.99 $\times$ , and the broadband marks (**void**, **signed**) actively *lose* information on  $\omega_c/n_s$  (down to 0.42 $\times$ ). The **signed** mark recovers a slight edge only on the amplitude/RSD parameters ( $\omega_b$  1.06 $\times$ ,  $\sigma_8$  1.19 $\times$ ). A scalar mark is a weighted average of pair counts; it cannot carry more cosmology than the unweighted  $\xi$  of the same population, and generally carries less.

**The gain was the legs, not the weight.** Splitting the 90-bin weighted vector shows the data-auto is the workhorse; the ASTRA-random auto and the cross alone are 2–4 $\times$  looser. Combining all three *does* tighten (**full** < **data** by  $\sim 1.3$ – $1.5\times$ ) — so the gain over the single quantile full-auto came from bundling three populations, precisely the “more legs” caveat, not from the weighting.

**Against the quantile multi-leg vectors, the weighting loses.** Compared not to one quantile statistic but to the project’s quantile *multi*-leg vectors (with their proper pooled 576-subbox covariance), the weighted vectors are beaten on every parameter:

Table 4: HOD-fixed marginalised  $\sigma$  at full-box volume. Quantile multi-leg vectors use the pooled 576-subbox covariance; weighted vectors the 192-sample *c000* reanalysis (so the weighted side is at a Hartlap *disadvantage*).

| vector                          | $n_b$ | $\omega_b$                             | $\omega_c$                             | $n_s$                                  | $\ln \sigma_8$                         |
|---------------------------------|-------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
| quantile full-auto              | 30    | $7.2 \times 10^{-3}$                   | $3.0 \times 10^{-3}$                   | $1.2 \times 10^{-3}$                   | $6.5 \times 10^{-3}$                   |
| quantile data-Q autos           | 120   | $2.4 \times 10^{-3}$                   | $1.8 \times 10^{-3}$                   | $5.1 \times 10^{-4}$                   | $4.1 \times 10^{-3}$                   |
| quantile full+data+rand Q autos | 270   | <b><math>1.1 \times 10^{-3}</math></b> | <b><math>1.2 \times 10^{-3}</math></b> | <b><math>3.6 \times 10^{-4}</math></b> | <b><math>3.0 \times 10^{-3}</math></b> |
| weighted rank full              | 90    | $5.6 \times 10^{-3}$                   | $1.7 \times 10^{-3}$                   | $7.2 \times 10^{-4}$                   | $5.0 \times 10^{-3}$                   |
| weighted signed full            | 90    | $4.3 \times 10^{-3}$                   | $3.5 \times 10^{-3}$                   | $1.5 \times 10^{-3}$                   | $4.0 \times 10^{-3}$                   |

The weighted-full gain versus the *best* quantile multi-leg is 0.19–0.75 $\times$  (i.e. the quantiles are 1.3–5 $\times$  tighter), worst on  $\omega_b$ . The earlier “weighted wins” held only against a single quantile statistic; the quantile splits retain the full conditional set  $\{\xi_{ab}\}$ , which is strictly more information than any scalar reweighting.

**What this means for the strand.** The continuous mark is a *compression* of the environment information, and as a compression it loses raw constraining power relative to keeping the quantile splits. Its value proposition is therefore *not* tighter Fisher errors but the secondary properties: lower dimensionality, differentiability/smoothness in the parameters (emulability), and the marked-statistics theory hook. Whether those buy anything in practice is the open question — an emulability head-to-head (weighted vs quantile vectors) is the test worth running before any larger weighted campaign.

## 11 Outlook

The continuous weights are a clean, validated environment compression, but on this grid the quantile multi-leg vectors carry more cosmological information; the mark’s case rests on emulability, not raw  $\sigma$ . Next are the HOD-clean ensemble Fisher (to remove the optimism above), an emulability test of the weighted vs quantile vectors, tunable exponents (the **power/exp** knobs), cross-weightings (void-weighted  $\times$  knot-weighted, the continuous analog of the void/knot cross legs that were the Fisher workhorses), and the standardised zero-mean mark.

## References

- [1] R. K. Sheth, “Marked correlations,” MNRAS **364** (2005) 796.
- [2] M. White, “Marked correlations and optimal weights,” JCAP **11** (2016) 057.
- [3] E. Massara et al., “Using the marked power spectrum to detect the signature of neutrinos,” PRL **126** (2021) 011301.